

**Ques. WHAT IS THE FREQUENCY BAND FOR TELEVISION CHANNEL BROADCASTS IN INDIA? IS THE SAME FREQUENCY USED FOR TV CHANNELS AND MOBILE COMMUNICATIONS ALL OVER THE WORLD? WHO DECIDES THE FREQUENCY SPECTRUM?**

**Rashmi Kulshrestha**

**Ans.** Television channels are available in VHF and UHF bands. Different countries use different frequency bandwidths and technologies. An international body, International Telecommunication Union (ITU) based in Geneva, provides for a unified television band

| GENERAL TELEVISION FREQUENCIES                  |                               |
|-------------------------------------------------|-------------------------------|
| Sub CATV Band - T7 - T13                        | 7 - 48 MHz                    |
| VHF Band - Ch. 2 - 13                           | 54 - 216 MHz                  |
| Low Band - VHF Ch. 2 - 6                        | 59 - 88 MHz                   |
| Mid Band - UHF Ch. 14 - 22<br>- UHF Ch. 95 - 99 | 121 - 174 MHz<br>91 - 120 MHz |
| High Band - VHF Ch. 7 - 13                      | 175 - 216 MHz                 |
| Super Band - CATV Ch. 23 - 36                   | 216 - 300 MHz                 |
| Hyper Band - CATV Ch. 37 - 62                   | 300 - 456 MHz                 |
| Ultra Band - CATV Ch. 63 - 158                  | 457 - 1002 MHz                |
| UHF Band Ch. 14 - 83 - CATV Ch. 63 - 158        | 70 - 1002 MHz                 |

Courtesy: [www.csgnetwork.com](http://www.csgnetwork.com)

plan, though not all countries observe this band plan. The ITU has been an intergovernmental public-private partnership organisation since its inception. India is one of ITU members. In India, all terrestrial TV spectrum allocation and broadcasting is controlled by the Ministry of Information and Broadcasting, Government of India.

The ITU coordinates the shared global use of the radio spectrum, promotes international cooperation in assigning satellite orbits, and works on worldwide technical standards. Different parts of the radio spectrum are allocated by the ITU for different radio transmission technologies and applications. Some general television frequencies are listed in the table above.

GSM frequency bands or frequency ranges are cellular frequencies designated by the ITU for GSM mobile phone operation. GSM-900 and GSM-1800 are used in most parts of the world, except countries like Japan and South Korea where GSM has never been introduced.

**Q2. WHAT IS A LOAD CELL? CAN WE INTERFACE IT WITH ARDUINO UNO R3 BOARD?**

**Pamarthi Kanakaraja**

**A2.** A load cell is a transducer that converts mechanical force into an electrical signal. The electrical signal can be voltage change, current change or frequency change depending on

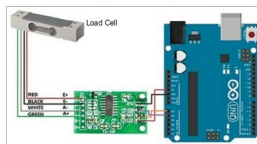


Fig. 1: Interfacing of a load cell with Arduino through an HX711 module

whether the load cell is resistive or capacitive. Load cells are normally used to measure force or weight. Strain-gauge load cells are commonly used.

You can interface a load cell with Arduino Uno R3 board. Fig. 1 shows interfacing of a load cell to an Arduino through an HX711 module.

**Q3. HOW IS A SPEAKER CONSTRUCTED? HOW DOES IT PRODUCE SOUND?**

**Sanket Sawalgi**

**A3.** A loudspeaker consists of paper or plastic moulded into a cone shape called 'diaphragm.' When audio signal is applied to the loudspeaker's voice coil suspended in a circular gap

between the poles of a permanent magnet, the coil moves rapidly back and forth due to Faraday's law of induction. This causes the diaphragm attached to the coil to move back and forth, pushing on the air to create sound waves.

Voice coil, usually made of copper wire, is glued to the back of the diaphragm. When sound signal passes through the voice coil, a magnetic field is produced around the coil causing the diaphragm to vibrate. The larger the magnet and voice coil, the greater the power and efficiency of the loudspeaker. The coil is oriented co-axially inside the gap; the outside of the gap being one pole and the centre post (called the pole piece) being the other.

The gap establishes a concentrated magnetic field between the two poles of the permanent magnet. The pole piece and backplate are often a single piece, called the pole plate or yoke, as shown in Fig. 2.

As pulses of electricity pass through the electromagnet coil, its magnetic field direction rapidly changes, causing the cone

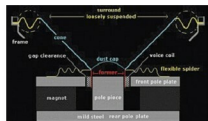


Fig. 2: Speaker construction

to vibrate. The cone structure amplifies these vibrations, pumping sound waves into the surrounding air and towards your ears. That's how sound comes from a loudspeaker.

Answers compiled by Sani Theo, technical editor.  
Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: [editsec@efy.in](mailto:editsec@efy.in)) and should include name and address of the sender